Dual-View Spectral and Global Spatial Feature Fusion Network for Hyperspectral Image Classification

Tan Guo, *Member, IEEE*, Ruizhi Wang, Fulin Luo, *Senior Member, IEEE*, Xiuwen Gong, Lei Zhang, *Senior Member, IEEE*, and Xinbo Gao, *Senior Member, IEEE*

***Abstract*— For hyperspectral image (HSI) classification, two branch networks generally use convolutional neural networks (CNNs) to extract the spatial features and long short-term memory (LSTM) to learn the spectral features. However, CNNs with a local kernel neglect the global properties of the whole HSI. LSTM does not consider the macroscopic and detailed information of spectra. In this article, we propose a dual-view spectral and global spatial feature fusion network (DSGSF) to extract the spatial–spectral features for HSI classification (HSIC), including a spatial subnetwork and a spectral subnetwork. In the spatial subnetwork, we propose a global spatial feature representation model based on the encoder–decoder structure with channel attention and spatial attention to learn the global spatial features. In the spectral subnetwork, we design a dual-view spectral feature aggregation model with view attention to learn the diversity of spectral features. By fusing the two subnetworks, we construct DSGSF to extract the spatial–spectral features of HSI with strong discriminating performance. Experimental results on three public datasets illustrate that the proposed method can achieve competitive results compared with the state-of-the-art methods. Code: https://github.com/RZWang-WH/DSGSF.**

***Index Terms*— Attention, encoder–decoder, global feature, hyperspectral image (HSI) classification, long short view feature.**

I. Introduction

HYPERSPECTRAL image (HSI) contains spatial information as well as rich spectral information, which can be represented as cubic data including two spatial dimensions and a spectral dimension [1]. When HSI is used for earth observation, it can accurately characterize the response of land covers. At present, HSI plays an important role in target detection [2], marine exploration [3], land planning [4], and many other fields [5]. For these applications, the fundamental problem is HSI classification (HSIC) to assign a semantic label to each pixel.

Deep-learning methods have been widely used to extract the high-level features of HSIC. For example, deep belief network (DBN) [6] was used for HSI feature extraction and classification. In [7], a stacked autoencoder (SAE) was used in an unsupervised manner to learn the intrinsic representation of hyperspectral data. In [8], a recurrent neural network (RNN) was used for HSIC by considering the spectral information as the sequence data. Hang et al. [9] proposed a cascaded RNN model to eliminate the redundant information and learn the complementary of non-adjacent spectra. These networks just consider the spectral information of each pixel, which neglects the spatial consistency of HSI.

To learn the spatial information of HSI, convolutional neural networks (CNNs) have been introduced to extract the high-level features of HSI and achieve effective performance for HSIC. 2-D-CNN [10] is the first application for HSIC to extract deep semantic features. He and Chen [11] optimized the input of CNNs by translating, rotating, and scaling the original input. In addition, CNN-based networks always lead to a drop in accuracy as the model goes deeper. To address this problem, residual networks and densely linked networks have been proposed to train deeper networks. By following the ideas, Zhong et al. [12] constructed a spectral–spatial residual network (SSRN) to extract spectral–spatial discriminative features of HSI. Wang et al. [13] proposed an end-to-end fast dense spectral–spatial convolution network (FDSSC) for HSIC. Yang et al. [14] embedded convolution operations into the transformer structure to capture subtle spectral differences.

The above studies show that both spatial and spectral features in deep learning play a key role in HSIC. With the development of spatial–spectral feature representation, the dual-branch structure [15], which separately extracts spatial and spectral features, has been consistently shown to be an effective framework for HSIC. To extract the spatial–spectral features under different scales, Cao and Guo [16] designed a cascaded dual-scale crossover network based on SSRN [12] for HSIC. Spectral–spatial unified network (SSUN) [17] uses

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Tan Guo and Ruizhi Wang are with the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: guot@cqupt.edu.cn; hswrzhi@163.com).

Fulin Luo is with the College of Computer Science, Chongqing University, Chongqing 400044, China (e-mail: luoflyn@163.com).

Xiuwen Gong is with the Faculty of Engineering, The University of Sydney, Camperdown, NSW 2006, Australia (e-mail: xiuwen.gong@sydney.edu.au).

Lei Zhang is with the School of Microelectronics and Communication Engineering, Chongqing University, Chongqing 400044, China (e-mail: leizhang@cqu.edu.cn).

Xinbo Gao is with the Chongqing Key Laboratory of Image Cognition, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: gaoxb@cqupt.edu.cn).

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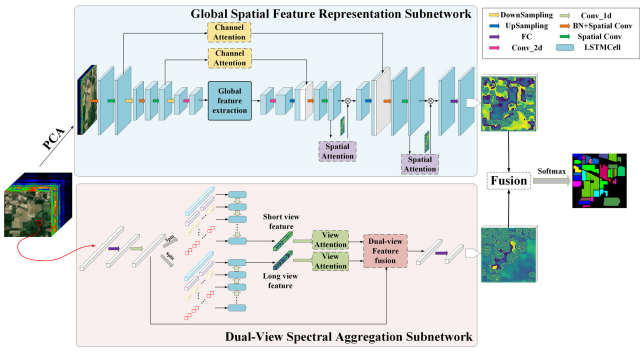


Fig. 1. Framework of the proposed DSGSF method.

a long short-term memory (LSTM) and a multiscale CNN to obtain the spectral–spatial fusion features. Yang et al. [18] designed a dual-branch deep CNN consisting of 1-D-CNN and 2-D-CNN to jointly extract spectral and spatial features from HSI. Wang et al. [19] proposed an adaptive spectral–spatial multiscale network (ASSMN), which uses the multiscale LSTM and convolution LSTM to extract the spectral and spatial features. Sun et al. [20] fused local spectral features and multiscale spatial features to mine the complementary information between the spatial and spectral features. Liu et al. [21] developed a fast dynamic graph convolutional network and CNN parallel network to extract the structural spatial–spectral features. And some methods [22], [23] based on graph neural networks have been proposed for learning discriminative spatial–spectral representations for HSI. Attention has played an important role in determining the importance of features and identifying key features. Recently, an attentional mechanism is widely used in HSIC [23]. Bai et al. [24] proposed a spectral–spatial transformer that assembles the spatial attention and extracts the spectral features in the HSI cube. Dong et al. [25] proposed the weighted fusion of CNNs and graph attention networks to address the problem of small samples for HSIC. To highlight the key features, the spatial attention and the channel attention [26] were introduced into the double-branch multiattention (DBMA) network [27] to extract more discriminative spatial–spectral features. To obtain the correlation between any two channels and any two pixels, the double-branch dual-attention (DBDA) network [28] designed the positional attention and the channel attention to enhance the feature dependencies. In [29], the self-attention module was introduced into the dual-branch structure to exploit the patch-based contextual information and the long-range spectral correlation. For the mentioned dual-branch structure, the spectral branch just learns the high-level spectral features, which does not consider the macroscopic and detailed information of spectra to reveal the trend and detail of the spectral curve. Meanwhile, the spatial branch usually divides HSI into some patches with fixed windows to learn the spatial features, which only focuses on the local information while ignoring the global relationships.

To address the above problems, we propose a dual-view spectral and global spatial feature fusion network (DSGSF) as shown in Fig. 1, including a global spatial feature representation subnetwork and a dual-view spectral aggregation subnetwork to learn the spatial and spectral information. The first subnetwork designs an encoder–decoder structure with a spatial convolution block composed of pointwise and depthwise convolution, and a global feature extraction module is embedded into the encoder–decoder structure to achieve the global spatial contextual information. In the second subnetwork, the dual-view can be likened to the process of changing the field of view caused by the telescoping of the microscope. The long view is the process of moving the view from near to far, and the short view is the process of moving the view from far to near. The two processes can extract the macro- and microinformation of features. For the dual-view features, we use the view attention based on the shared features to calculate the weight coefficients of features in a lightweight way. The contributions of this article are listed as follows.

1) We propose an end-to-end network named DSGSF by using the global spatial representation and the dual-view spectral information to capture the discriminative features of HSI. Compared with existing spatial–spectral joint networks, our network has richer and more efficient feature representation.
2) A global spatial feature representation subnetwork based on the encoder–decoder structure is proposed. In the subnetwork, channel attention is used to enhance the connection of low- and high-level semantics, and spatial attention is used to enhance spatial feature representation. This network learns the low-level, high-level, and global spatial features, which overcomes the limitation of the receptive field (RF) to reveal the global contextual spatial features of HSI.
3) A dual-view spectral aggregation subnetwork that fuses spectral information from different views is proposed to improve spectral discrimination. This subnetwork uses the macro- and microinformation views to refine the spectral features. In addition, spectral attention is used to enhance the spectral features from different views.

The rest of this article is organized as follows. Section II gives the details about the proposed method. Section III shows the experimental results and discusses the effectiveness of the network. In Section IV, we draw some conclusions.

II. Methodology

A. Overall Framework

Fig. 1 shows the general framework of the proposed DSGSF. The global spatial feature representation subnetwork is used as a spatial subnetwork to extract the spatial representation and the dual-view spectral aggregation subnetwork is used as a spectral subnetwork to extract the spectral representation. Specifically, $X \in \mathbb{R}^{H\times W\times B}$ represents the original HSI data, where $H \times W$ is the spatial size and B is the band number of HSI. Since the similarity between bands is very strong, we employ principal component analysis (PCA) to reduce the dimensionality of HSI before feeding it into the spatial subnetwork. The reduced HSI can be denoted as $I \in \mathbb{R}^{H\times W\times b}$, where b is the reduced dimension by PCA and $b \ll B$.

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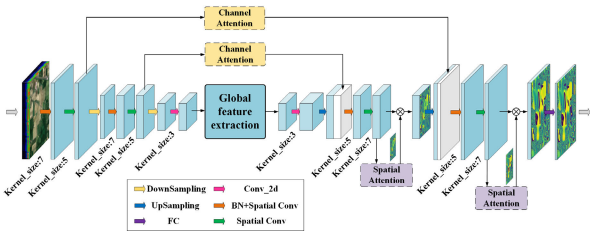


Fig. 2. Global spatial feature representation subnetwork.

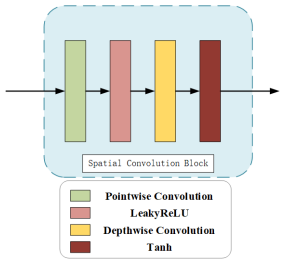


Fig. 3. Spatial convolution block.

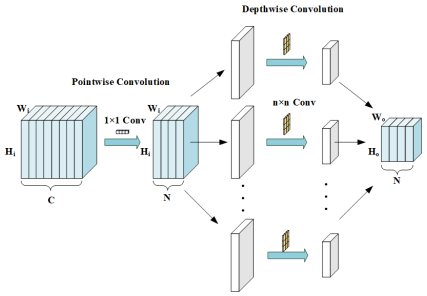


Fig. 4. Structures of pointwise and depthwise convolution.

To make full use of the spatial information, I is used as an input to the spatial subnetwork based on the encoder–decoder structures with spatial attention and channel attention. The spatial features $F_{\text{sa}} \in \mathbb{R}^{H\times W\times L}$ can be produced, where L represents the number of land-cover categories. To perceive the discriminative spectral characteristics, a pixel $p_i \in \mathbb{R}^{1\times 1\times B}$ is taken as the input of the spectral subnetwork. The long view and short view features are extracted by the spectral aggregation module composed of LSTM cells, and its representation features are enhanced by the view attention. Then, the spectral feature $F_{\text{se}} \in \mathbb{R}^{1\times 1\times L}$ can be produced. Finally, the spatial features and spectral features are fused by means of "score weighting".

B. Global Spatial Feature Representation Subnetwork

1) Architecture Design: In the convolutional network, the RF is of great significance for feature representation, and a larger RF can extract features with more global and higher semantic levels. In the previous research, to obtain a larger RF, there are mainly the following three methods. The first is to directly increase the size of the convolution kernel, but this operation is bound to increase a large number of parameters. The second is to deepen the network, but this approach comes with the risk of vanishing gradients. The third is to use pooling to reduce the feature map, which will lead to the loss of feature information. To avoid the above shortcomings, we use the encoder–decoder structure network to expand the RF, as shown in Fig. 2. This subnetwork includes five modules, namely the spatial convolutional block, 2-D-CNN, the spatial attention module, the channel attention module, and the global feature extraction module.

When the encoder–decoder structure is downsampling, some feature information will inevitably be lost, and the lost features cannot be learned from the upsampling. To make the feature expression more complete, the skip connection is adopted. The shallow features and high-level features are spliced and passed backward, and channel attention is used to enhance the shallow features before splicing. After each skip connection, spatial attention is used to enhance important features and suppress meaningless information.

2) Spatial Convolution Block: In the network, each downsampling reduces the size of the feature map to 1/4, and the smallest feature map is only 1/16 of the size of the original HSI. We designed a spatial convolution block as shown in Fig. 3, containing a depthwise convolution and a pointwise convolution. And the structures of pointwise and depthwise convolution are shown in Fig. 4. Some researchers [30] believe that the depthwise and pointwise convolution combined with activation function and batch normalization can improve the nonlinear expression ability of the network, thereby making the network more powerful.

For the spatial convolution block, we use the depthwise convolution after the pointwise convolution and add a nonlinear activation function in the middle to enhance the mapping relationship of features. Then, a nonlinear activation function is added to the tail of the block to improve the nonlinear expression ability of the block.

Suppose the input feature map size is $H_i \times W_i \times C$, the convolution kernel is $D \times D \times N$, where N represents the number of kernels, and the output feature map size is $H_o \times W_o \times N$. For a single 2-D-CNN, the computational amount is

$$H_i \times W_i \times D \times D \times C \times N. \tag{1}$$

Using a similar analysis for a spatial convolution block, the computational amount is

$$C \times N \times H_i \times W_i + H_o \times W_o \times D \times D \times N. \tag{2}$$

The number of parameters for a single 2-D-CNN is

$$C \times N \times D \times D. \tag{3}$$

The number of parameters for a spatial convolution block is

$$N \times D \times D + C \times N. \tag{4}$$

Compared with an ordinary convolution, the ratio of the computational amount of a spatial convolution block to an ordinary convolution is $(1/D^2) + ((H_o \times W_o)/(C \times H_i \times W_i))$, and the ratio of the parameters amount is $(1/C) + (1/D^2)$.

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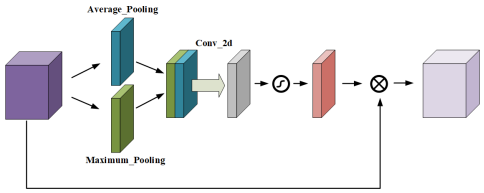


Fig. 5. Spatial attention module.

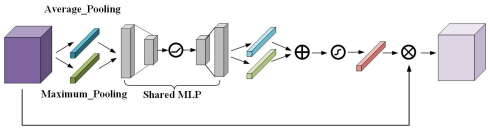


Fig. 6. Channel attention module.

3) Spatial Attention: Spatial attention aims to enhance the spatial difference of pixels from different classes. And the weight matrix of spatial attention has the same size as the input feature map. Fig. 5 shows spatial attention.

The input feature maps are first processed with global average and maximum pooling. Then, the two outputs are concatenated horizontally as the input of the convolutional layer with an activation function to obtain the attention weight matrix. Finally, we multiply the input feature map with the weight matrix

$$s_{\text{avg}} = \frac{1}{C}\sum_{c=1}^{C} f_c \tag{5}$$

$$s_{\max} = \max_{c}\ (f) \tag{6}$$

$$A_{\text{sa}} = f_c \otimes \sigma([s_{\text{avg}}, s_{\max}] * W) \tag{7}$$

where f_c is the Cth channel of the input feature map $f \in \mathbb{R}^{H\times W\times C}$. s_{avg} and $s_{\max}$ are the results of average pooling and maximum pooling on the channel dimension. W refers to the weight of the convolution, $[\cdot]$ represents concatenation, and $\sigma(\cdot)$ donates the sigmoid function.

4) Channel Attention: The skip connections are used to enhance the relevance between the encoding information and the decoding information. However, the extracted low-level features have a lot of redundant information. Soft attention can effectively suppress the features in irrelevant regions and reduce the redundant information for skip connections. In this article, we use channel attention [26] to enhance the skip connections. Fig. 6 shows the channel attention.

Let the input feature map be $f \in \mathbb{R}^{H\times W\times C}$, and CBAM [26] indicates that the global maximum pooling is considered as the complementary of global average pooling

$$s_{\text{avg}} = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} f(i,j) \tag{8}$$

$$s_{\max} = \max_{\text{sa}}\ (f) \tag{9}$$

where s_{avg} and $s_{\max}$ are the results of average pooling and maximum pooling on the spatial dimension, respectively.

To enhance the difference of channel features and preserve low-level responses of negative outputs, LeakyReLU is added

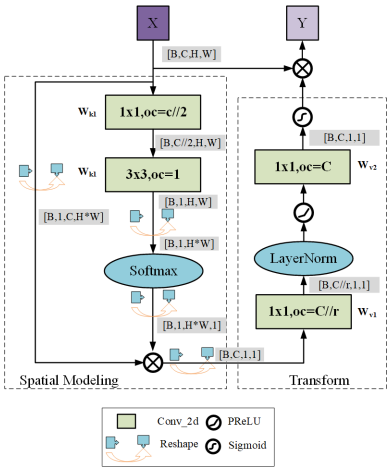


Fig. 7. Global feature representation module.

to the bottleneck with two MLP layers. In addition, the dimensionality reduction rate is r

$$h = (s, W) = (W_2, \delta(W_1, s)) \tag{10}$$

where $W_1 \in \mathbb{R}^{C\times(C/r)}$ and $W_2 \in \mathbb{R}^{(C/r)\times C}$ are the wights of the two MLP layers, $\delta(\cdot)$ refers to LeakyReLU, and the reduction ratio r is set to 16 in experiments.

To reduce the number of parameters, the parameters W_1 and W_2 are shared for the outputs of maximum and average pooling. Then, two outputs h_{avg} and $h_{\max}$ are added together to multiply with the original data after a nonlinear mapping

$$h = h_{\text{avg}} + h_{\max} \tag{11}$$

$$\begin{aligned} A_{\text{se}} &= f \otimes \sigma(h) \\ &= f \otimes \sigma((W_2, \delta(W_1, s_{\text{avg}})) + (W_2, \delta(W_1, s_{\max}))). \end{aligned} \tag{12}$$

5) Global Feature Representation Module: For HSIC, the input data is divided into multiple patches, which is beneficial to detect edge and texture features. But this process will lose some dependencies with other pixels, especially the similarity between pixels from the same class. Therefore, we consider the dependencies between all pixels of the entire feature map to extract global features. The global feature representation module is shown in Fig. 7.

In this module, to reduce the number of parameters, the bottleneck structure is adopted, where r is the reduction rate. In addition, adding normalization before ReLU can reduce the difficulty and improve the generalization ability. The module main includes spatial modeling, transform, and feature fusion.

For the spatial modeling stage, it is used to obtain the weights of the global features and strengthen the spatial features. The weight matrix is obtained by the correlation coefficients between the current position and the other positions, and the global features are continuously strengthened as the number of training increases. The weighted global features O_s of x_i are expressed as follows:

$$O_s = \sum_{i=1}^{H*W} \frac{e^{W_{k2}W_{k1}x_i}}{\sum_{m=1}^{H*W} e^{W_{k2}W_{k1}x_m}} x_i \tag{13}$$

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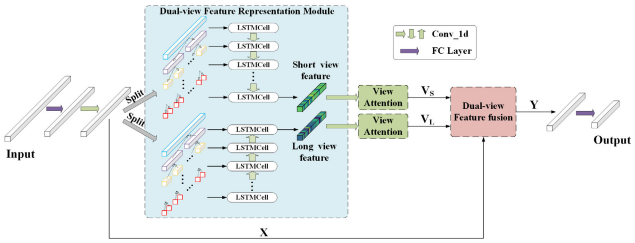


Fig. 8. Dual-view spectral aggregation subnetwork.

where $W_{k1} \in \mathbb{R}^{C\times(C/2)}$ and $W_{k2} \in \mathbb{R}^{3\times3\times(C/2)}$ are the weight parameters.

The transform stage is used to extract the interchannel dependencies. In this article, a bottleneck structure based on 1×1 convolution is used to transform the global features as channel dependencies with the weights between interchannels. And a ReLU activation function is employed before normalization. The transform features O_t are represented as

$$O_t = \sigma(W_{v2}(\delta(\mathrm{LN}(W_{v1} * O_s))) \tag{14}$$

where $W_{v1} \in \mathbb{R}^{C\times(C/r)}$ and $W_{v2} \in \mathbb{R}^{(C/r)\times C}$ are the weight parameters, $\mathrm{LN}(\cdot)$ is the layer normalization, $\sigma(\cdot)$ is the sigmoid function, and $\delta(\cdot)$ denotes the LeakyReLU function.

For the feature fusion stage, the feature fusion method is used to learn the global features of each position, which is represented as

$$Y = X \times O_t. \tag{15}$$

In summary, the spatial modeling part uses the entire feature as input, and the RF covers the entire feature. However, the process of convolution is a local operation that only obtains a restricted RF. In addition, simple convolution stacking is considered as the fitting features with a function in the same form. The stacked convolutions make the features lack diversity, while the global feature representation module by fusing spatial modeling and transformation increases the diversity of features.

C. Dual-View Spectral Aggregation Subnetwork

1) Architecture Design: For the previous LSTM based on spectral information, the reflectance values of each band are sequentially input into the LSTM [31], which will generate long-term dependencies [32]. In the training process, this manner will suffer the problem of gradient disappearance or gradient explosion. Thus, some research works have tried to use spectral vector grouping and multiscale strategies in LSTM to reduce the effects of long-term dependence [17], [19]. However, in this strategy, the features at different scales are independent. Therefore, it is necessary to consider the relationships between different scale features to achieve a more natural fusion of the variables between different scales. Based on this, we design a dual-view spectral aggregation subnetwork as shown in Fig. 8.

The proposed subnetwork consists of a dual-view feature representation module, two view attention modules, a dual-view feature fusion module, and some 1-D-CNNs. The dual-view feature representation module learns the long-view and short-view features. Then, the view attention module is used to enhance the dual-view features. Finally, the enhanced dual-view features are fused by the dual-view feature fusion module together with the original features. This subnetwork provides a complete description of the macroscopic and detailed information of the spectral features.

2) Dual-View Feature Representation Module: To process the spectral information, we divide the spectral features into multiple features with different lengths, which can be regarded as multiscale features. At the same time, we use two directions, including from the large scale to the small scale and from the small scale to the large scale, to observe the multiscale features, which are considered dual-view information. Based on the dual-view information, we use the LSTM cell to construct a dual-view feature representation module, as shown in Fig. 8.

For this module, the inputs for both long and short views are the same multiscale vectors that are divided by halving the feature vectors consecutively. In the short-view feature representation process, it passes the features from the large scale to the small scale. The hidden and cell states are transformed to the next smaller scale by 1-D CNN with the next scale features as the input of the next LSTM cell. The feature transfer process is represented in the following:

$$\left(h_t^i, c_t^i\right) = \mathrm{LSTMCell}\left(x_t^i, \left(W_h^{\lceil i/2\rceil} * h_{t-1}^{\lceil i/2\rceil}, W_c^{\lceil i/2\rceil} * c_{t-1}^{\lceil i/2\rceil}\right)\right) \quad i \in (1, \ldots, 2^t) \tag{16}$$

where $\mathrm{LSTMCell}(\cdot)$ represents the LSTM cell operation, and (h_t^i, c_t^i) denotes the output of the hidden state and the cell state corresponding to the ith sequence at the tth scale. x_t^i is the input state of the ith sequence at the tth scale, $\lceil\cdot\rceil$ represents the operation of round up, $W_h^{\lceil i/2\rceil}$ and $W_c^{\lceil i/2\rceil}$ are the parameters of 1-D-CNN to the hidden and cell states for the sequence $\lceil i/2\rceil$ at the previous scale $t-1$, respectively, $h_{t-1}^{\lceil i/2\rceil}$ and $c_{t-1}^{\lceil i/2\rceil}$ are the output hidden and cell states of the previous scale $t-1$, respectively.

For the long-view feature representation, the hidden and cell states are passed from the small-scale to the large-scale by the transformation of 1-D-CNN, and then they are input into the next LSTM cell with the next scale features. In the processes, the 1-D-CNNs are used to adjust the scale of the hidden and cell features for matching the scale of each LSTM input state. The feature transfer process is represented as

$$\left(h_t^i, c_t^i\right) = \mathrm{LSTMCell}\left(x_t^i, \left(W_h^i * \left[h_{t+1}^{i*2-1}, h_{t+1}^{i*2}\right], W_c^i * \left[c_{t+1}^{i*2-1}, c_{t+1}^{i*2}\right]\right)\right), \quad i \in (1, \ldots, 2^t) \tag{17}$$

where $[h_{t+1}^{i*2-1}, h_{t+1}^{i*2}]$ and $[c_{t+1}^{i*2-1}, c_{t+1}^{i*2}]$ denote the concatenation operation of two adjacent sequences corresponding to the output hidden and cell states of the previous scale $t+1$, respectively, W_h^i and W_c^i are the parameters of 1-D-CNN to the hidden and cell states for the concatenated sequence corresponding to the current sequence i.

Based on the above feature transfer process, the output hidden state of the smallest scale is considered the short-view feature, and the output cell state of the largest scale is regarded as the long-view feature.

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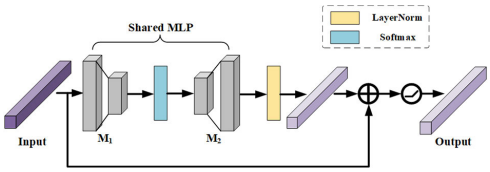


Fig. 9. View attention module.

3) View Attention Module: Specifically, the dual-view spectral aggregation subnetwork learns the multiscale features that have some redundancy. Therefore, the intrinsic connection of features is used to enhance effective information and suppress redundant and invalid information. To address this problem, we design a view attention to highlight the key information, as shown in Fig. 9.

For view attention, a shared MLP and a normalization layer are used to represent the attention features, where the shared MLP includes two linear layers and a softmax operation. The proposed view attention is a simple module with a very low computational cost, in which the additive manner is used to represent the attention operation. Let the input view features be $V_{\text{in}} \in \mathbb{R}^{1\times d}$, and the view attention features is represented as

$$V_{\text{out}} = \delta(V_{\text{in}} \oplus \text{LN}(\text{Softmax}(V_{\text{in}}M_1)M_2)) \tag{18}$$

where V_{out} represents the output of the attention features, $\delta(\cdot)$ is the LeakyReLU function, "$\oplus$" is the additive operation for using the attention to the view features, LN$(\cdot)$ is the layer normalization, Softmax$(\cdot)$ is the operation of softmax, and M_1 and M_2 are the parameters of two linear layers.

The view attention modules are used for the short-view and the long-view features, respectively, and obtain the enhanced-view features V_S and V_L. Since this attention utilizes the additive operation and the linear transformation, the computational complexity and the number of parameters are not excessive.

4) Dual-View Feature Fusion: To associate the long-view and short-view features, a layer of LSTM is directly used as a feature aggregation module. To preserve the original spectral information, the original features are used as the input of LSTM. Therefore, in the fused LSTM module, the long-view features are used as the cell state, the short-view features as the hidden state, and the original features as the input state. The dual-view features fusion can be represented as

$$Y, _ = \text{LSTMCell}(X, (V_S, V_L)) \tag{19}$$

where LSTMCell$(\cdot)$ represents an LSTM cell, X refers to the feature of the original spectrum after transformation, (V_S, V_L) is the short- and long-view features, and Y denotes the output of the dual-view feature fusion module.

D. Feature Fusion and Loss Function

Based on the two subnetworks, we get two outputs with the same number of channels, where the number of channels is the number of categories. To fuse the outputs of the different subnetworks, we flatten the spatial features. And the output spatial features are selected by the indexes of the pixels corresponding to the spectral features so that the spatial features and the spectral features have the same shape. We set the output of the flattened spatial subnetwork to be $F_{\text{sa}} \in \mathbb{R}^L$ and the output of the spectral subnetwork to be $F_{\text{se}} \in \mathbb{R}^L$, where L represents the number of categories.

The features from different domains have different discriminating performances. In addition, different domain features have different contributions for different categories. If the fusion method is unreasonable to represent the contribution of different domain features, the fusion features will generate a negative impact on classification. Therefore, we adopt the adaptive "score weighting" method to fuse the features of two subnetworks. This method can adaptively emphasize the important features when fusing the two subnetworks' features. The obtained features are more flexible and discriminative. The fusion manner is

$$O = f(\gamma)F_{\text{sa}} + (1 - f(\gamma))F_{\text{se}} \tag{20}$$

where γ is a learnable parameter with a random initial value, which can adaptively reveal the contribution of different domain features. $f(\cdot)$ represents the activation function Sigmoid to convert the weights into the range of (0, 1).

Since the range of the output value of the network is uncertain, the softmax function is used to convert the output value of the network into the range of (0, 1), which denotes the probability of the features belonging to a certain category as follows:

$$p_{ij} = \frac{e^{O_{ij}}}{\sum_{j=1}^{C} e^{O_{ij}}}. \tag{21}$$

Then, the multitask cross-entropy loss is used for the optimization of DSGSF, and it can be denoted as

$$Ls = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij} \log p_{ij} \tag{22}$$

where Ls represents the loss value, N is the number of training samples, C is the number of categories, p_{ij} refers to the predicted probability value that the ith sample belongs to the jth class, and y_{ij} is the truth label indicator

$$y_{ij} = \begin{cases} 1, & j = l_i \\ 0, & j \neq l_i \end{cases} \tag{23}$$

where l_i is the true label information of the ith sample.

III. Experiments and Result Analysis

A. Data Description

In the experiments, we use three HSI datasets to evaluate the effectiveness of the proposed method, including Indian Pines (IN), Pavia University (PU), and Salinas (SA) as shown in Fig. 10.

1) Indian Pines: IN, the common test dataset for HSIC, was imaged by the Airborne Visible Infrared Imaging Spectrophotometer (AVIRIS) in Indiana, USA, including 200 bands with the size of 145 × 145 pixels and 20-m spatial resolution. It contains 16 types of ground objects. The wavelength range of this dataset is 0.4–2.5 μm. In each class, we randomly

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Ground Truth	Class ID	Color	Class Name	Train	Val	Test	Total
Indian Pines	1		Alfalfa	10	3	33	46
	2		Corn-notill	72	15	1341	1428
	3		Corn-mintill	42	9	779	830
	4		Corn	12	3	222	237
	5		Pasture	25	5	453	483
	6		Trees/Grass	37	8	685	730
	7		Pasture-mowed	10	3	15	28
	8		Hay-windrowed	24	5	449	478
	9		Oats	10	3	7	20
	10		Soybeans-notill	49	10	913	972
	11		Soybean-mintill	123	25	2307	2455
	12		Soybeans-cleantill	30	6	557	593
	13		Wheat	11	3	191	205
	14		Woods	64	13	1188	1265
	15		Building-Grass	20	4	362	386
	16		Stone-steelTowers	10	3	80	93
	Total			549	118	9582	10249
Pavia University	1		Asphalt	67	67	6497	6631
	2		Meadows	187	187	18275	18649
	3		Granel	21	21	2057	2099
	4		Trees	31	31	3002	3064
	5		Painted metal sheets	14	14	1317	1345
	6		Bare Soil	51	51	4927	5029
	7		Bitumen	14	14	1302	1330
	8		Self-Blocking Bricks	37	37	3608	3682
	9		Shadows	10	10	927	947
	Total			432	432	41912	42776
Salinas	1		Brocoli_green_weeds_1	21	21	1967	2009
	2		Brocoli_green_weeds_22	38	38	3650	3726
	3		Fallow	20	20	1936	1976
	4		Fallow_rough_plow	14	14	1366	1394
	5		Fallow_smooth	27	27	2624	2678
	6		Stubble	40	40	3879	3959
	7		Celery	36	36	3507	3579
	8		Grapes_untrained	113	113	11045	11271
	9		Soil_vinyard_develop	63	63	6077	6203
	10		Corn_senesced_green_weeds	33	33	3212	3278
	11		Lettuce_romaine_4wk	11	11	1046	1068
	12		Lettuce_romaine_5wk	20	20	1887	1927
	13		Lettuce_romaine_6wk	10	10	896	916
	14		Lettuce_romaine_7wk	11	11	1048	1070
	15		Vinyard_untrained	73	73	7122	7268
	16		Vinyard_vertical_trellis	19	19	1769	1807
	Total			549	549	53031	54129

Fig. 10. Detailed information of three datasets for experiments.

select 5%, 1%, and 94% of the samples for training, validation, and testing, respectively. The detailed information is shown in Fig. 10.

2) Pavia University: The PU data was captured by the Reflective Optical System Imaging Spectrometer (ROSIS) sensor in 2003. The HSI has 103 bands in the wavelength range of 0.43–0.86 μm, and the spatial resolution is 1.3 m. Nine categories are included in this data, as shown in Fig. 10, which details the randomly selected 1%, 1%, and 98% samples for training, validation, and testing, respectively.

3) Salinas: The SA data was also captured by the AVIRIS sensor, which contains the SA Valley areas, in California, USA. It has a spatial resolution of 3.7 m and contains 204 bands. The size of this image is 512 × 217, and it contains a total of 54 129 samples, which are divided into 16 categories in total. The detailed information is shown in Fig. 10. 1%, 1%, and 98% of samples for each class are randomly selected for training, validation, and testing, respectively.

B. Experimental Setting

To verify the performance, we conducted a series of experiments on the three datasets. And three evaluation indicators were used to measure the classification performance, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC). OA is the ratio of the correct labels to the total number of labels, AA is the average accuracy of all the classes, and the kappa coefficient measures the consistency between the predicted labels and the ground truth. We also analyzed the complexity with the number of parameters (Params) and floating-point operations (FLOPs). All the experiments were implemented on Ubuntu 20.04 and a GPU of NVIDIA RTX 3090 with 24 GB RAM.

For a fair comparison, all the HSIC frameworks were implemented on Pytorch. We used the Adam optimizer to update the training parameters. The epoch of training was fixed to 200. We used the model with the best performance on the validation set to evaluate the test set. We randomly split the labeled samples ten times, and all the results were the average of the ten repeated experimental results. Both the average value and the standard deviation were reported. Taking the reduced dimensionality of PCA into account, we selected 0.99 principal components of variation on the three datasets.

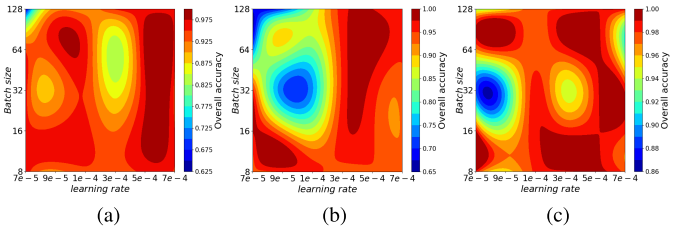


Fig. 11. Performance of DSGSF with different learning rates and batch size on the three datasets. (a) IN. (b) PU. (c) SA.

C. Hyperparameter Analysis

We analyzed the effect of the proposed method on the overall accuracy with respect to the learning rate and batch size. Fig. 11 shows the OAs under different parameters of learning rate and batch size. The regions with high classification accuracies are mainly located on the right side of the figures. In addition, DSGSF has a lower impact on classification accuracy on the SA dataset. In this article, the learning rate and batch size are set to 0.0005 and 64, respectively.

D. Compare With Other Methods

In this section, the three datasets were used to evaluate the classification performance of the proposed model. We compared our method with other existing state-of-the-art methods, such as support vector machine (SVM) [33], bi-directional LSTM (Bi-LSTM) [34], going deeper with contextual CNN (CDCNN) [35], residual spectral–spatial attention network (RSSAN) [36], band-adaptive spectral–spatial feature learning neural network (BASSNet) [37], SSRN [12], FDSSC [13], ASSMN [19], DBMA [27], and DBDA [28]. Specifically, SVM is the classical machine-learning method, CDCNN extracts only spatial information and Bi-LSTM extracts only spectral features. SSRN, FDSSC, RSSAN, BASSNet, DBDA, DBMA, ASSMN, and DSGSF are all deep-learning methods based on the fusion of spatial and spectral features. In these methods, RSSAN and BASSNet are serial networks to extract the spatial–spectral features, implying a progressive extraction of the spatial–spectral features for HSIC. SSRN and FDSSC are 3-D-CNN-based methods to jointly extract the spatial–spectral features. DBDA, DBMA, ASSMN, and DSGSF are all parallel two-branch networks, where each branch extracts spatial and spectral features. The classification results of different methods on each dataset are shown in Tables I–III, and the corresponding classification maps are shown in Figs. 12–14.

1) Quantitative Evaluation: From Tables I–III, most deep-learning methods achieve better performance than SVM on the three datasets. SVM relies on raw or shallow features for classification, while deep learning can extract more abstract

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TABLE I

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET

Class	SVM	Bi-LSTM	CDCNN	RSSAN	BASSNet	SSRN	FDSSC	DBMA	DBDA	ASSMN	DSGSF
1	37.36±11.64	43.41±8.05	45.15±28.17	97.09±3.94	89.80±9.62	97.66±3.15	97.97±2.93	96.67±2.72	97.78±2.08	96.52±6.52	**99.40±1.21**
2	63.68±3.31	67.50±4.34	65.13±12.65	80.87±1.89	76.31±4.45	94.82±1.81	94.47±2.62	92.27±2.99	96.06±1.94	93.90±3.74	**97.78±0.71**
3	68.16±3.27	62.46±5.02	56.34±10.50	80.86±4.63	69.04±4.63	95.01±2.83	94.69±3.11	93.04±2.17	95.71±1.16	96.48±2.42	**97.38±0.71**
4	52.88±7.22	53.67±8.12	56.24±30.58	58.76±9.53	49.40±11.05	92.39±4.24	91.26 ±5.71	77.45±11.36	86.22±8.25	94.95±4.37	**97.94±0.80**
5	86.82±4.50	84.26±4.40	93.34±5.76	83.981±5.80	88.66±5.64	92.41±3.21	93.21 ±3.10	91.09±3.48	93.35±4.07	97.15±1.91	**98.45±0.95**
6	87.62±2.41	89.39±2.33	89.15±8.64	90.11±5.97	98.75±0.96	98.75±1.21	99.59 ± 0.43	97.75±1.92	**99.62±0.20**	96.79±1.88	99.37±0.51
7	56.45±9.43	62.25±13.50	45.40±22.53	96.16±5.15	98.89±2.22	**100.00±0.00**	**100.00 ± 0.00**	**100.00±0.00**	**100.00 ± 0.00**	88.40±13.67	89.83±7.71
8	95.43±1.64	94.40±1.03	94.11±4.99	99.26±0.85	99.35±0.56	99.62±0.52	99.87 ± 0.20	98.79±1.15	99.82±0.35	99.78±0.31	**100.0±0.00**
9	30.02±6.28	43.28±8.94	39.14±19.22	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	90.59±15.30	93.36±13.34
10	69.10±3.89	67.50±4.76	65.83±12.48	83.79±3.93	79.64±5.04	91.25±3.63	91.06±3.45	89.42±3.80	90.52±4.19	95.39±2.12	**96.75±0.94**
11	71.05±2.06	72.03±2.46	70.44±12.90	92.56±2.02	87.45±4.20	96.65±1.75	96.70±0.92	93.96±1.37	96.34±1.29	94.30±2.04	**99.15±0.19**
12	61.87±5.18	53.65±6.79	51.08±8.91	72.945±5.72	60.78±8.12	94.49±2.19	94.61±1.99	94.11±1.21	97.34±1.36	90.89±3.77	**97.82±0.65**
13	87.15±4.48	90.94±4.94	92.46±8.74	99.20±1.90	99.16±0.42	99.37±0.65	98.74±1.63	98.64±0.98	**100.00 ± 0.00**	99.54±0.57	**99.79±0.34**
14	89.78±1.57	89.31±1.90	89.38±6.71	96.06±1.41	97.07±1.61	98.60±1.29	**98.70±1.03**	97.10±1.82	98.30±1.35	97.13±2.26	97.31±0.55
15	68.88±6.80	68.00±4.65	74.85±7.40	76.81±1.41	71.46±8.00	92.41±4.01	94.37 ± 2.99	91.66±2.60	93.26±3.00	94.24±5.60	**99.39±0.54**
16	96.87±3.31	81.15±7.82	90.75±5.83	76.578±17.72	98.64±1.70	96.91±3.95	**97.65±2.94**	99.75±0.48	**99.76±0.48**	96.72±3.07	95.92±0.92
OA(%)	74.22±1.39	73.74±0.85	72.02±7.67	86.81±1.58	83.60±1.35	95.76±0.78	95.85±0.68	93.67±2.86	96.02±0.86	95.25±0.76	**98.24±0.12**
AA(%)	70.20±1.53	70.20±1.58	69.92±8.04	86.57±2.38	85.27±1.34	96.27±0.84	96.43±0.64	94.48±1.90	96.51±0.82	95.17±1.44	**97.48±1.02**
KC×100	70.37±1.58	69.96±1.00	67.61±9.48	84.90±1.82	81.18±1.52	95.16±0.89	95.26±0.79	92.78±0.99	95.45±0.98	94.57±0.87	**97.99±0.14**
Train time (s)	17.4±0.45	14.42±1.21	31.21±0.83	101.72±6.10	90.4±2.55	66.58±3.25	145.3±7.25	34.53 ±23.04	44.06±5.11	480.53±45.85	164.23±11.20
Test time (s)	1.18±0.05	1.27±0.15	1.85±0.32	3.63±1.35	2.82±2.13	2.34±0.67	5.37±2.23	2.94±1.53	5.96±1.20	3.05±1.09	1.68±0.68
Params(k)	-	163.22	1063.57	64.21	111.53	370.47	2414.78	381.53	606.10	3558.05	403.53
FLOPs(M)	-	2.62	112.64	80.52	206.71	6181.43	13589.52	1013.67	2425.83	30241.02	889.58

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PU DATASET

Class	SVM	Bi-LSTM	CDCNN	RSSAN	BASSNet	SSRN	FDSSC	DBMA	DBDA	ASSMN	DSGSF
1	88.31±1.77	82.41±4.28	83.06±5.96	87.51±3.32	91.36±2.97	**99.49±0.54**	97.66±1.26	96.37±2.18	98.07±1.32	90.39±2.44	99.12±0.34
2	91.59±1.54	89.72±0.83	90.36±5.07	98.06±0.84	98.80±0.74	**99.71±0.53**	98.98±0.47	98.10±0.63	99.44±0.47	93.53±2.12	99.46±0.19
3	71.04±4.74	48.12±7.20	44.16±26.62	52.59±6.86	60.30±8.81	94.22±2.82	85.55±5.96	94.00±5.37	81.04±9.70	80.56±5.86	**98.22±0.21**
4	93.90±1.78	87.98±3.64	97.65±1.37	90.84±4.88	94.36±1.96	97.54±1.05	94.04±2.59	**99.04±0.73**	94.48±1.77	97.88±1.78	98.70±0.52
5	96.42±2.57	97.49±1.71	97.75±1.95	97.84±1.29	99.61±0.52	**100.00±0.00**	99.50±0.93	99.92±0.11	99.96±0.60	99.11±1.05	99.90±0.04
6	85.90±4.67	82.68±3.44	85.07±7.07	88.68±6.23	79.63±4.97	99.09±1.23	98.70±1.00	97.90±1.22	98.68±1.24	89.65±6.50	**99.54±0.30**
7	75.84±7.20	55.21±8.30	43.91±36.35	59.91±5.28	72.41±10.14	**98.46±2.41**	92.44±3.84	98.03±1.27	94.53±3.20	72.10±9.91	**98.46±1.25**
8	79.95±3.05	69.83±3.60	69.57±6.21	76.24±4.51	82.26±6.38	**97.82±1.10**	90.37±5.49	86.93±3.89	96.22±2.11	95.38±2.63	97.19±0.61
9	99.95±0.05	99.18±0.43	95.97±3.69	91.32±2.66	99.67±0.27	**99.94±0.07**	99.28±0.28	99.64±0.34	99.78±0.12	97.68±3.29	99.67±0.16
OA(%)	88.37±0.55	83.64±0.85	85.76±3.61	89.36±1.73	90.99±0.78	98.99±0.42	98.85±0.48	96.80±0.58	97.47±0.75	91.87±1.70	**99.09±0.20**
AA(%)	87.00±0.74	79.18±1.80	78.61±8.22	82.55±2.17	86.49±1.57	98.47±0.57	98.07±0.67	95.12±0.62	95.80±0.98	90.70±2.01	**98.92±0.37**
KC×100	84.43±0.76	78.07±1.11	80.70±5.14	85.81±2.32	87.95±1.04	98.66±0.55	98.47±0.64	95.77±0.78	96.64±1.37	89.13±2.27	**98.79±0.27**
Train time (s)	6.09±0.21	11.34±1.82	31.5±5.45	90.79±6.14	71.05±9.23	47.89±5.26	91.62±8.21	45.09±7.08	52.63±8.01	376.79±64.28	205.55±17.74
Test time (s)	2.84±0.14	1.19±0.09	5.74±0.95	9.21±1.01	12.85±3.13	9.96±1.63	21.73±5.41	9.30±1.20	15.09±3.24	13.31±4.21	2.48±0.56
Params(k)	-	41.86	628.11	33.40	41.76	216.54	1235.46	201.96	320.69	3462.03	389.635
FLOPs(M)	-	0.68	60.56	47.28	62.89	3129.76	6826.07	654.13	1447.19	30237.73	9951.43

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SA DATASET

Class	SVM	Bi-LSTM	CDCNN	RSSAN	BASSNet	SSRN	FDSSC	DBMA	DBDA	ASSMN	DSGSF
1	99.69±0.32	98.52±1.64	37.10±43.28	89.72±7.03	99.43 ± 0.85	99.40 ± 0.96	99.92 ± 0.16	99.71±0.43	98.79±1.48	98.44±3.07	**99.96±0.06**
2	98.95±0.41	97.79±1.28	68.29±17.26	88.70±4.71	99.46 ± 0.49	99.99 ± 0.02	**100.00 ± 0.00**	99.98±0.02	99.11±1.79	99.28±0.59	99.92±0.09
3	91.15±2.85	92.18±1.88	87.57±9.80	79.77±10.98	94.86 ± 4.48	99.25 ± 1.23	99.42 ± 1.08	97.90±2.01	98.69±1.21	99.54±0.52	**99.66±0.66**
4	97.77±0.64	96.57±1.20	94.08±4.07	90.27±5.21	99.22 ± 0.68	**99.63 ± 0.42**	**99.63 ± 0.18**	99.22±0.68	98.25±0.46	95.26±1.68	98.74±1.63
5	96.29±2.49	97.81±0.97	96.40±5.96	96.84±2.22	97.94 ± 1.43	**99.46 ± 0.42**	99.31 ± 0.69	95.80±0.91	98.00±8.44	98.05±1.23	99.32±0.61
6	99.90±0.12	99.80±0.20	98.84±0.82	96.70±1.27	99.64 ± 0.50	**100.00 ± 0.00**	**100.00 ± 0.00**	99.95±0.57	**100.00±0.00**	99.09±1.58	99.69±0.80
7	97.96±1.12	97.20±1.50	95.18±4.58	89.29±9.82	99.37 ± 0.81	99.95 ± 0.05	99.91 ± 0.08	99.93±0.64	**99.98±0.14**	99.61±0.56	99.80±0.26
8	75.72±2.53	74.56±1.62	73.51±10.77	86.06±3.08	82.10 ± 6.10	92.88 ± 1.80	93.57 ± 1.69	87.89±8.42	84.20±9.70	89.91±8.03	**99.48±0.49**
9	98.63±0.30	98.58±0.60	95.66±4.93	96.62±1.59	99.46 ± 0.45	**99.95 ± 0.13**	99.98 ± 0.07	99.91±0.16	99.49±1.02	97.58±2.81	99.88±0.23
10	88.58±2.35	87.56±3.16	80.40±11.67	82.27±5.11	92.52 ± 2.05	97.06 ± 0.97	96.56 ± 1.24	94.96±2.48	95.91±2.06	94.86±2.69	**98.73±0.44**
11	89.87±4.81	85.55±3.56	73.58±27.10	83.44±4.02	88.18 ± 9.20	97.66 ± 2.53	97.41 ± 3.08	95.71±5.03	97.06±2.27	93.03±6.58	**99.32±1.03**
12	96.52±1.13	95.29±1.77	84.26±12.33	89.38±4.79	99.39 ± 1.62	**100.00 ± 0.00**	99.99 ± 0.02	99.97±0.25	**100.00 ± 0.00**	98.21±2.37	98.42±3.28
13	94.30±2.23	95.19±1.38	76.31±38.31	95.81±1.94	**99.75 ± 0.31**	99.68 ± 0.48	99.70 ± 0.29	98.56±0.93	98.61±0.93	90.51±8.99	97.98±1.86
14	94.46±3.25	95.62±1.09	94.38±7.70	88.09±6.83	95.04 ± 1.59	98.04 ± 1.36	98.58 ± 1.14	94.69±4.04	98.17±0.96	94.14±3.81	**99.02±0.84**
15	76.05±3.90	70.40±2.24	57.64±23.22	86.12±5.92	73.61 ± 7.15	91.07 ± 2.69	90.98 ± 2.71	74.14±17.74	86.28±8.07	89.70±4.58	**98.86±0.16**
16	97.97±1.07	96.73±1.01	98.43±1.07	83.89±9.66	97.09 ± 1.58	97.76 ± 1.74	98.30 ± 1.76	97.84±1.47	97.82±1.59	99.10±1.37	99.90±0.25
OA(%)	89.42±0.62	88.24±0.40	79.94±8.18	90.55±2.34	91.32±1.08	96.88±0.26	97.02±0.40	93.07±2.79	94.07±1.36	94.68±2.76	**99.37±0.32**
AA(%)	93.36±0.35	92.46±0.33	81.98±9.45	89.63±2.86	94.82±0.69	98.24±0.31	98.33±0.35	96.01±1.58	96.90±0.68	96.02±1.42	**99.23±0.27**
KC×100	88.20±0.70	86.88±0.45	77.65±9.06	87.41±2.62	90.34±1.19	96.53±0.29	96.68±0.45	92.28±3.15	93.41±1.49	94.05±3.10	**99.29±0.36**
Train time (s)	12.21±0.36	15.19±2.34	40.76±3.21	121±18.96	140.77±26.83	87.81±9.20	180.88±7.99	49.61±3.75	52.06±4.30	483.47±55.51	185.98±15.85
Test time (s)	4.43±0.16	1.10±0.88	7.17±0.39	13.38±3.02	9.97±2.14	10.4±2.10	21.62±4.68	11.80±3.01	31.98±4.77	17.51±3.53	1.85±0.38
Params(k)	-	169.74	1081.49	64.70	95.24	384.31	2463.60	388.83	617.72	3561.63	403.35
FLOPs(M)	-	2.74	114.79	81.88	207.73	6241.94	13869.05	1034.29	2475.64	30241.12	7123.55

high-level semantics. Both Bi-LSTM and CDCNN ignore the interconnection between the spatial and spectral information of HSI, so they show insufficient feature extraction ability with a small number of samples, and the performance results are worse than SVM. For the deep-learning-based methods, SSRN and FDSSC extracting jointly spatial–spectral features

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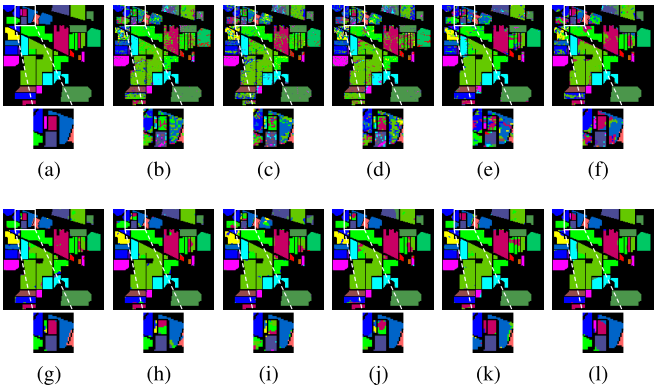


Fig. 12. Classification maps on the IN dataset. (a) Ground truth. (b) SVM. (c) Bi-LSTM. (d) CDCNN. (e) RSSAN. (f) BASSNet. (g) SSRN. (h) FDSSC. (i) DBMA. (j) DBDA. (k) ASSMN. (l) DSGSF.

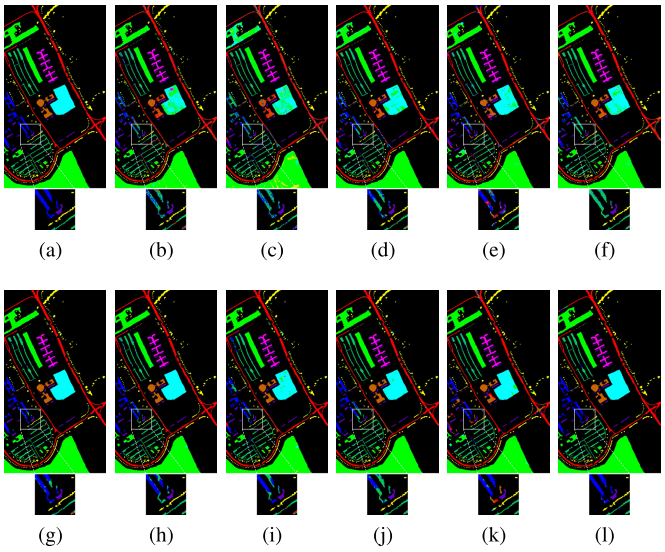


Fig. 13. Classification maps on the PU dataset. (a) Ground truth. (b) SVM. (c) Bi-LSTM. (d) CDCNN. (e) RSSAN. (f) BASSNet. (g) SSRN. (h) FDSSC. (i) DBMA. (j) DBDA. (k) ASSMN. (l) DSGSF.

perform better than RSSAN and BASSNet learning serially spatial–spectral features. The two-branch networks (i.e., DBDA, DBMA, ASSMN, and DSGSF) outperform RSSAN and BASSNet. For the two-branch networks, they have no significant advantages over the networks that jointly extract spatial–spectral features, and vice versa, while the proposed network has the best results in these networks. Although serial networks are also effective to extract spectral and spatial features, the spectral or spatial features will be corrupted when transmitting from the low-level layers to the high-level layers. Therefore, the performance of the serial networks is weaker than the other methods based on the joint spatial–spectral features. The proposed DSGSF method has more advantages for different scenes because it not only considers the global spatial relationship, but also makes full use of the macroscopic structure and microscopic details of spectra. Especially, on the SA dataset, DSGSF outperforms the best of the comparison methods (i.e., SSRN and FDSSC) over 2.49% and 2.35%, respectively.

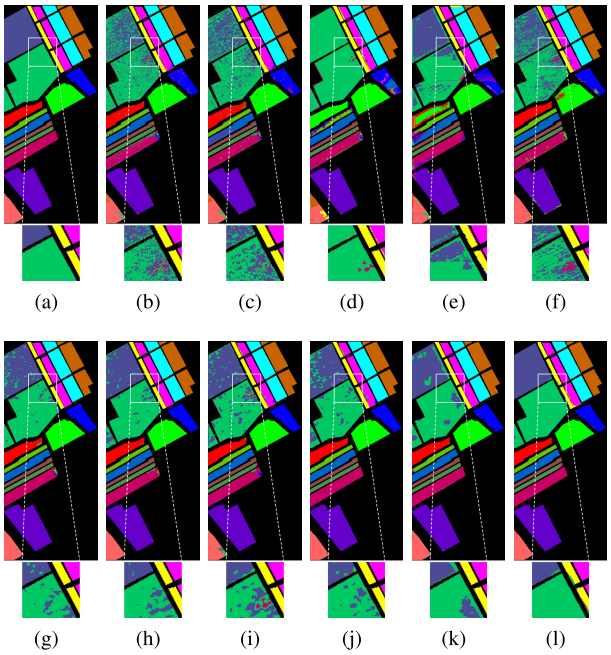


Fig. 14. Classification maps on the SA dataset. (a) Ground truth. (b) SVM. (c) Bi-LSTM. (d) CDCNN. (e) RSSAN. (f) BASSNet. (g) SSRN. (h) FDSSC. (i) DBMA. (j) DBDA. (k) ASSMN. (l) DSGSF.

2) Qualitative Evaluation: The classification maps of these methods are shown in Figs. 12–14. By comparing real ground reference and classification maps, the classification results of SVM and Bi-LSTM contain a large amount of noise. Looking at all the classification results on the IN and SA datasets, the misclassified pixels are mainly concentrated in the upper part of the image, and on the PU dataset, many pixels are misclassified in the middle of the image. For a closer look at the classification result maps, we enlarged a local part of the classification map, which contains multiple categories. DSGSF is more accurate in the local enlarged map than the compared methods. Specifically, on the SA dataset, the classification performance of DSGSF has a definite advantage due to the reason that DSGSF can represent the global spatial features and macro–micro dual-view features. In general, the proposed DSGSF has the best visual classification performance compared with the other methods under a limited number of training samples.

For a more intuitive view of the performance of each method, we used t-SNE on the PU dataset to visualize the feature discriminative performance as shown in Fig. 15. The interclass margin of the results of RSSAN and BASSNet is very small, and there is a large number of confounded samples from different classes. The interclass margins of the results of SSRN, DBDA, DBMA, and DSGSF are wider, while the intraclass distribution of DBDA and DBMA is also larger. The distribution results of SSRN and DSGSF are similar, but there are more mixed samples for SSRN than DSGSF. In addition, DSGSF shows the best classification distribution performance compared with all the other methods.

3) Time and Complexity Analysis: To evaluate the running time and complexity of the proposed method and the other state-of-the-art methods, the complexity of deep learning and

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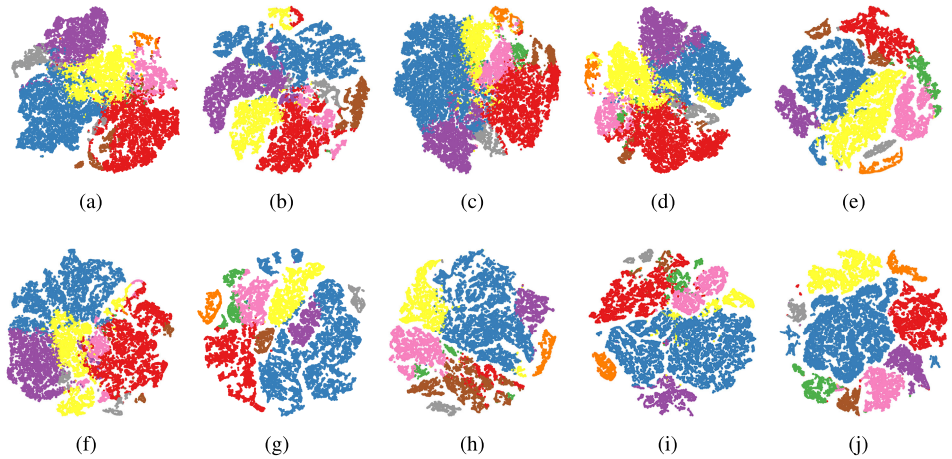


Fig. 15. Visualization with t-SNE on the PU dataset. (a) Bi-LSTM. (b) CDCNN. (c) RSSAN. (d) BASSNet. (e) SSRN. (f) FDSSC. (g) DBMA. (h) DBDA. (i) ASSMN. (j) DSGSF.

the time cost are shown in Tables I–III. The training time of each model is mainly determined by a combination of FLOPs and the training epochs, which represents the model fitting time. From the fitting time, it is not difficult to see that the simpler models cost less for training than the complex model. According to the previous works [32], we know that the time cost of LSTM is relatively large, which is affected by the structure of the LSTM model itself. Therefore, the training time of ASSMN and DSGSF is longer than the other methods. The number of parameters in the models shows that the simpler the structure, the lower the complexity of the model. In the serial models, CDCNN has the highest complexity because of its deeper network. In the two-branch models, SSRN, DBMA, and DBDA have low complexity due to their simple structures, while FDSSC has higher complexity due to its dense links. The complexity of ASSMN is the highest among the compared methods because it uses ConvLSTM with multiple time-step calculations and multibranch architectures. DSGSF has a low complexity among the two-branch models due to its lightweight spatial convolutional block and view attention module.

TABLE IV
OAs (%) OF THE ABLATION ON THE SPATIAL SUBNETWORK

Module	IN	PU	SA
Spa	96.36±2.08	97.34±1.24	98.08±0.86
Spa+g	97.56±1.06	97.54±1.02	98.65±0.61
Spa+sa	97.45±1.17	97.37±1.16	98.60±0.68
Spa+ca	97.35±1.32	97.64±0.92	98.46±0.77
Spa+sa+ca	97.40±0.88	97.96±0.53	98.68±0.60
Spa+g+sa	97.53±0.69	98.08±0.45	98.55±0.70
Spa+g+ca	97.67±0.63	97.57±0.99	98.63±0.44
Spa+g+sa+ca	97.71±0.42	98.23±0.35	98.88±0.36

TABLE V
OAs (%) OF THE ABLATION ON THE SPECTRAL SUBNETWORK

Module	IN	PU	SA
Spe	73.53±1.02	84.13±0.75	87.88±1.05
Spe+sv	73.95±0.79	85.04±0.53	88.64±0.87
Spe+lv	73.63±0.86	84.55±0.69	88.32±0.26
Spe+lv+sv	75.62±0.37	86.01±0.44	88.97±0.31
Spe+lv+sv+va	77.48±0.33	88.87±0.21	90.27±0.17

E. Discussion and Analysis

1) Ablation for Spatial Subnetwork: The spatial subnetwork contains the base spatial network (Spa) only with skip connection encoder–decoder structure, the global feature representation module (g), the spatial attention module (sa), and the channel attention module (ca). We compared the performance with and without these modules on different datasets as shown in Table IV.

According to Table IV, any of these modules can improve the performance of the spatial subnetwork. There are different performances on each dataset. For example, on the IN dataset, the combination of the spatial attention module and the channel attention module performs weaker than the spatial attention module alone, because this dataset has complex and small regions for each category to impact the representation of spatial attention. However, the classification accuracy of the spatial subnetwork with all three modules significantly improves. These results show that the three modules can improve the spatial subnetwork to extract more effective spatial discriminative features.

2) Ablation for Spectral Subnetwork: The spectral subnetwork mainly consists of the base spectral network (Spe) (it directly inputs the spectral features into one LSTM cell), the long-view module (lv), the short-view module (sv), and the view attention module (va). We compared with or without these modules on the spectral subnetwork. The experimental results are shown in Table V.

Since the local features can effectively reflect more detailed spectral properties, the short-view module generates better performance than the long-view module. Specifically, after adding the view attention module, the spectral subnetworks are improved by 1.86%, 2.86%, and 1.3%, respectively. In addition, the long- and short-view information with the view attention enhances the discriminating power of features to effectively improve the classification performance of the spectral subnetwork.

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TABLE VI
OAs (%) OF THE ABLATION ON THE OVERALL NETWORK

Module	IN(%)	PU(%)	SA(%)
Spa-sub	97.71±0.42	98.23±0.35	98.88±0.36
Spe-sub	77.48±0.33	88.87±0.21	90.27±0.17
DSGSF	98.24± 0.12	99.09±0.20	99.37±0.32

TABLE VII
IMPACT OF DIFFERENT CONVOLUTION STRATEGIES ON THE SPATIAL SUBNETWORKS

Module	IN(%)	PU(%)	SA(%)	Params (k)	FLOPs(M)
Spa-blk	97.71	98.23	98.88	194.54	885.13
2D-blk	92.48	93.87	90.27	1108.35	23276.82

3) Ablation for Overall Network: Table VI shows the results on the three datasets for the overall network and each branch. The results show that the overall network outperforms the performance of each branch, which demonstrates the effectiveness of spatial–spectral feature fusion. In addition, the spatial subnetwork (Spa-sub) performs better than the spectral subnetwork (Spe-sub), but the spatial and spectral features can complement each other.

4) Training Sample Size Analysis: Considering the stability and robustness of the proposed method under different numbers of training samples, 2%, 4%, 6%, 8%, and 10% labeled samples were randomly selected as training data on the IN dataset and 0.6%, 0.8%, 1.0%, 1.2%, and 1.4% on the PU and SA datasets. In this experiment, we selected the deep-learning methods based on spatial–spectral fusion for comparison, and five experiments were conducted for each case. We used the average OA to describe the accuracies under different training sample sizes for each model. The results are shown in Fig. 16.

In Fig. 16, the OAs of all the methods improve with the increase of training samples. The poor performance of BASSNet and RSSAN on all three datasets is due to the fact that BASSNet and RSSAN extract spatial and spectral information in tandem. The tandem process of different operations will lose the learned features. Compared with the other seven methods, DSGSF performs the best results under different training sample ratios on the three datasets. Especially, the advantages of DSGSF are particularly evident in the datasets with relatively simple spatial distribution. For example, on the SA dataset, the performance of DSGSF maintains an absolute advantage with different sample sizes.

5) Different Convolution Strategies Analysis: To verify the effectiveness of the spatial convolution block, we compared the effect of the spatial convolution block (Spa-blk) and the 2-D convolution block (2-D-blk) on the global spatial feature representation subnetwork as shown in Table VII.

In Table VII, the spatial convolution block achieves better results than the 2-D convolution block on the three datasets, which indicates that the spatial convolution block can enhance the diversity of convolution to improve the discriminative performance of features. In addition, we also compared the computational cost under different convolution strategies, and Params and FLOPs are in Table VII. The results show that the spatial convolution block reduces the complexity of Params and FLOPs compared with the 2-D convolution block.

TABLE VIII
COMPLEXITY OF THE PROPOSED NETWORKS ON THE IN DATASET

Complexity	direct g	Spa-sub w/o g	Spa-sub	Spe-sub w/o va	Spe-sub	DSGSF
Params (k)	194.54	191.62	194.54	208.95	208.99	403.53
FLOPs (M)	4053.58	881.93	885.13	4.38	4.45	889.58

TABLE IX
RESULTS OF DIFFERENT METHODS ON THE OTHER DATASETS (%)

		SSRN	FDSSC	DBDA	DBMA	DSGSF
WLK	OA	94.72±1.52	95.47±2.12	96.85±2.34	93.96±2.27	**98.00±0.55**
	AA	82.07±4.81	84.22±6.81	88.06±9.60	80.47±5.55	**95.92±0.44**
	KC×100	93.03±1.99	94.03±2.66	95.83±3.11	91.99±2.99	**97.49±0.73**
H18	OA	83.34±2.00	84.89±2.79	87.18±1.52	86.38±1.25	**87.74±0.22**
	AA	71.14±9.38	72.62±7.90	79.31±5.91	81.89±4.26	**82.27±0.84**
	KC×100	79.49±2.68	80.24±3.63	83.29±2.00	82.27±1.62	**84.03±0.30**

6) Complexity of Model Analysis: To validate the complexity of the proposed network, we analyzed the computational resources to be consumed by the spatial and spectral subnetworks on the IN dataset. We calculated the cost to directly input the original HSI into the global feature represent module without any other modules (direct g). We also gave the complexity of the spatial subnetwork without (w/o) global feature represent module (that is the encoder–decoder structure), the spatial subnetwork, the spectral subnetwork without view attention, the spectral subnetwork, and the DSGSF network, which are shown in Table VIII.

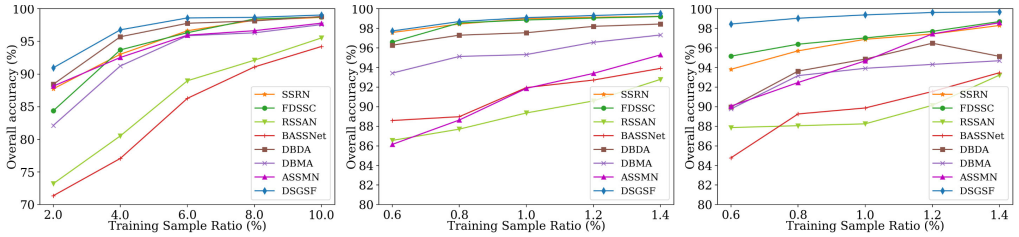
In Table VIII, for the spatial subnetworks, the direct global feature representation module requires a very large computational cost, but embedding the global feature representation module into the encoder–decoder structure greatly reduces the amount of computation. While these models based on the spatial subnetwork have a similar number of parameters. For the spectral subnetwork, the view attention requires the calculation of only two memory cells, so the increased amount of computation and the number of parameters is very low. Furthermore, the spectral subnetwork has a larger number of parameters than the spatial subnetwork, but it requires less computation than the spatial subnetwork.

F. Experimental Results on the Other Datasets

To further analyze the effectiveness of the proposed method, we validated several advanced models and the proposed model on two other datasets. These methods were validated on the WHU-HI-LONGKOU (WLK) with 550 × 400 pixels including nine categories of land covers and Houston2018 (H18) with 2384 × 601 pixels including 20 categories of land covers. We randomly select 0.2%, 0.2%, and 99.6% samples from the WLK dataset, and 0.5%, 0.5%, and 99% samples from the H18 dataset for training, validating, and testing, respectively. All the results are the average of ten repeated experimental results reported in Table IX.

The results show that DBDA outperforms the other comparison methods on the two datasets. However, the proposed DSGSF outperforms DBDA over 1.25% on the WLK dataset, and over 0.56% on the H18 dataset in terms of OA. The

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(a) (b) (c)

Fig. 16. OAs (%) of DSGSF with different training sample ratios on three datasets. (a) IN. (b) PU. (c) SA.

experimental results show that the proposed DSGSF has good performance on large-scale datasets.

IV. Conclusion

In this article, we present a DSGSF for HSIC including the spatial subnetwork (i.e., global spatial feature representation) and the spectral subnetwork (i.e., dual-view spectral aggregation). The spatial subnetwork represents the global structure relationship and enhances the discriminative power of spatial features. For the spectral subnetwork, it constructs the dual-view feature representation to reveal the macroscopic and detailed properties of spectra. DSGSF not only reveals the global structure properties based on the encoder–decoder structure and global spatial association, but also enhances the discrimination and diversity of features with the attention and dual-view information. The proposed method needs labeled information to train the network. In the future, we will explore constructing semisupervised learning [38], which can use the unlabeled samples for training to reduce the cost of labeling samples.

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Tan Guo (Member, IEEE) received the M.E. degree in signal and information processing and the Ph.D. degree in communication and information systems from Chongqing University, Chongqing, China, in 2014 and 2017, respectively.

He was a Post-Doctoral Fellow with the Macau University of Science and Technology, Taipa, Macau, China. Since 2018, he has been with the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, China. His research interests include computer vision, pattern recognition, and machine learning.



Ruizhi Wang received the B.E. degree in communication engineering from Huaibei Normal University, Huaibei, China, in 2021. He is currently pursuing the M.E. degree in electronics and communications engineering with the Chongqing University of Posts and Telecommunications, Chongqing, China.

His research interests include hyperspectral image classification, remote-sensing image processing, and deep learning.



Fulin Luo (Senior Member, IEEE) received the B.E. degree in mechanical engineering and automation from Southwest Petroleum University, Chengdu, China, in 2011, the M.E. and Ph.D. degrees in instrument science and technology from Chongqing University, Chongqing, China, in 2013 and 2016, respectively.

He was an Associate Researcher and a Post-Doctoral Researcher with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, from 2017 to 2021. He was a Research Fellow with Nanyang Technological University, Singapore, from 2020 to 2021. He has been a Professor with the College of Computer Science, Chongqing University, since 2022. His research interests include remote-sensing processing, computer vision, and biomedical analysis.



Xiuwen Gong received the B.E. and M.E. degrees from Anhui Normal University, Wuhu, China, in 2011 and 2014, respectively. She is currently pursuing the Ph.D. degree in artificial intelligence from the Faculty of Engineering, The University of Sydney, Camperdown, NSW, Australia.

Her research interests include machine learning, deep learning, and artificial intelligence (AI).



Lei Zhang (Senior Member, IEEE) received the Ph.D. degree in circuits and systems from the College of Communication Engineering, Chongqing University, Chongqing, China, in 2013.

He was a Post-Doctoral Fellow with The Hong Kong Polytechnic University, Hong Kong, from 2013 to 2015. He is currently a Professor/Distinguished Research Fellow with Chongqing University. His research interests include machine learning, pattern recognition, computer vision, and intelligent systems.



Xinbo Gao (Senior Member, IEEE) received the B.Eng., M.Sc., and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, China, in 1994, 1997, and 1999, respectively.

From 1997 to 1998, he was a Research Fellow at the Department of Computer Science, Shizuoka University, Shizuoka, Japan. From 2000 to 2001, he was a Post-Doctoral Research Fellow at the Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong. Since 2001, he has been at the School of Electronic Engineering, Xidian University. He is currently a Professor of control science and engineering with Xidian University and the President of the Chongqing University of Posts and Telecommunications, Chongqing, China. His research interests include computer vision, machine learning, pattern recognition, artificial intelligence, and smart city.

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